1 A ball rolls forwards and backwards on a curved track as shown in Fig. 1.1.

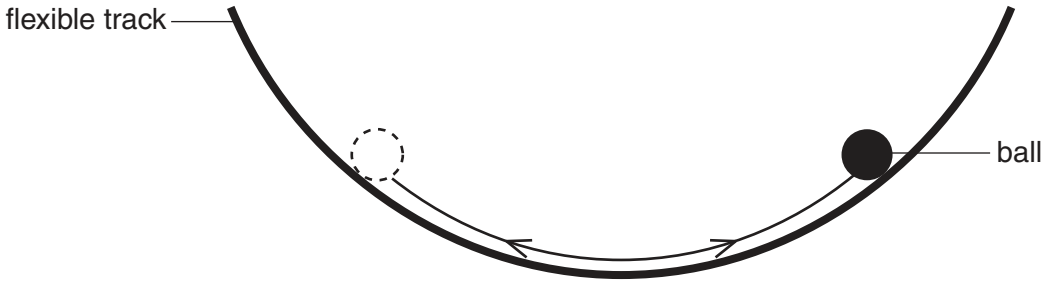

Fig. 1.1

It is suggested that the period T of the oscillations is related to the radius r of the ball and the radius of curvature C of the track by the relationship

$$T^2 = \frac{28\pi^2}{5g}(C - r)$$

where g is the acceleration of free fall.

You are provided with a flexible track. Design a laboratory experiment to test the relationship between T and r. Explain how your results could be used to determine a value for C. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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